

THE ARCHER

Adjustable Basketball Training Stick

Designed & Developed by Matthew D. Fogarty

Product Overview

The Archer is a patent-protected, adjustable-height training tool that creates a physical obstacle players must shoot over, forcing proper arc, higher release points, and game-realistic shot mechanics. Built from a weighted base, telescopic aluminum pole (adjustable from 4 to 18 feet), a rotatable joint adapter, and a bendable crossbar, the entire unit is portable, durable, and usable on any court surface, indoors or outdoors.

Why I Built It

Flat shots are one of the most common and fixable problems in basketball. Players at every level lose efficiency because they push the ball forward instead of lifting it through shooting mechanics and arc. Existing training aids are either fixed to the rim, too expensive for most programs, or limited to one drill. Before The Archer, I was holding a stick or raising my arm in front of shooters to force a higher arc. That gets tiring fast, and it locks me in one spot instead of letting me move around the player to observe their footwork, balance, elbow alignment, and follow-through. I needed something that could hold that position for me so I could actually coach. I wanted a tool that was portable, adjustable, and versatile enough to address arc, release point, rhythm, and defensive simulation all in one, and that freed me up to teach, not just stand there. So I built it.

How It Works

The Archer is placed between the shooter and the basket at varying distances and heights depending on the training objective. The pole adjusts from 6 to 18 feet, allowing coaches to match the height to the specific drill, shot type, or player need. Two primary placements drive development:

Close Placement (2–10 feet from the shooter): Forces a higher release point, corrects flat or pushed shots, and reinforces the ground-up power chain. Legs load first, the ball

rides the lift, and the wrist flips at the top. This is the primary tool for fixing shooting mechanics.

Halfway Placement (midpoint between shooter and rim): Simulates a real defender's contest at the point where the ball's arc should peak. Trains game-realistic trajectory, deep-range consistency, and natural power transfer from legs to release. This is the primary tool for game-ready shooting.

Development Areas

Skill Area	How The Archer Develops It
Shooting Arc	Eliminates flat shots by forcing a 45–50° trajectory over the obstacle
Release Point	Demands full arm extension and a high release to clear the crossbar
Shooting Rhythm	Reinforces the correct down-up chain: legs load and lead, the ball rides the lift from pocket to set point, and the arm extends with a wrist flip at the top. Prevents players from arming the ball by teaching ground-up power transfer.
Leg Power Transfer	Forces players to generate lift from their legs instead of pushing with their arms
Contested Shots	Simulates a defender's hands-up contest at game-realistic distances
Deep-Range Consistency	Maintains proper arc at 20–25+ feet by training correct shot peak at the midpoint
Floater & Runners	Adjustable height allows training for touch shots in the lane over length

Results With My Players

I developed and introduced The Archer during player development sessions at Santa Ana College, using it consistently across individual workouts, team shooting drills, and pre-practice skill work. The results were immediate and measurable:

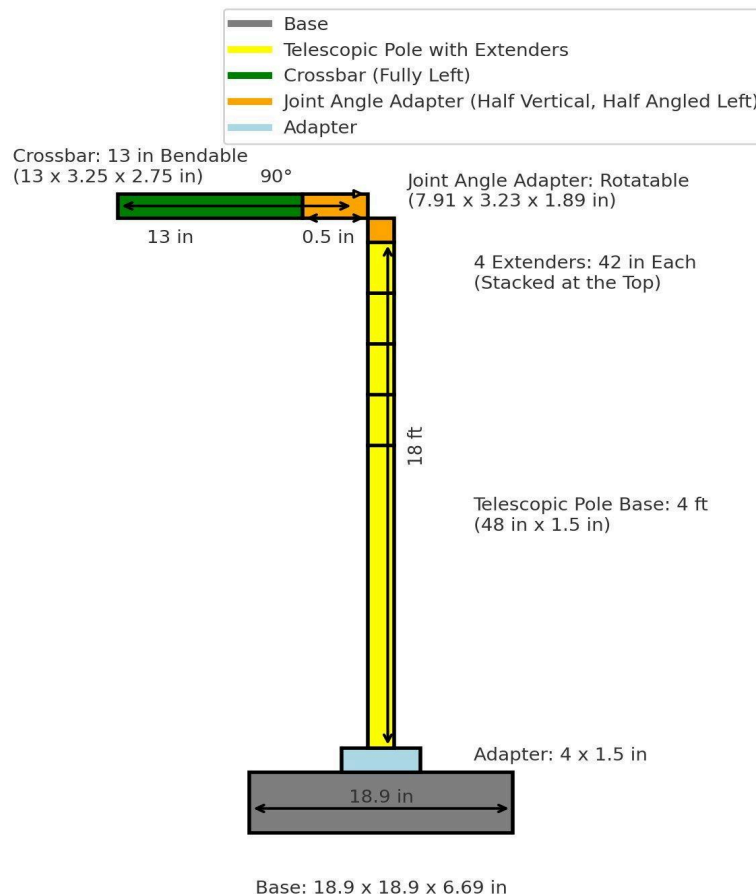
- Players who consistently shot flat or pushed the ball forward began producing proper arc within the first week of use. The visual and physical feedback from The Archer corrected habits that verbal coaching alone had not fixed.
- Shooting percentages from mid-range and three-point range improved as players developed softer touch and more consistent trajectory. Shots that

previously hit hard off the back rim began dropping with backspin and better angle.

- Players reported increased confidence shooting over length and in contested situations. The repetition of clearing the crossbar translated directly to composure when defenders closed out in games.
- Rhythm and timing improved across the board. The Archer naturally reinforced the correct ground-up sequencing, and players began generating power from their legs rather than their upper body. This reduced fatigue and eliminated front-rim misses late in games.
- Younger and less experienced players showed the fastest improvement. The tool gave them an external reference point that accelerated form correction far more efficiently than traditional drill repetition alone.

Product Specifications

Blueprint: Adjustable Training Stick (Crossbar Angles Shown)



Component	Details
Weighted Base	18.9 x 18.9 x 6.69 in — heavy-duty plastic, 8.8 lbs, fillable with sand/water
Adapter Reducer	4 in length — 201 stainless steel, connects pole to base
Telescopic Pole	4–18 ft adjustable — aluminum with ABS foam grip, four 42 in extending sections
Joint Angle Adapter	7.91 x 3.23 x 1.89 in — ABS plastic, rotatable and lockable at 90°
Crossbar	13 in bendable — flexible rubber with microfiber padding, anti-scratch, washable

Vision & Direction

The Archer started as a solution to a problem I was dealing with every day on the court. It has since become a core piece of my player development system and a tool I plan to continue refining and expanding. The next steps include producing a standardized version with custom-manufactured components, building a training curriculum around it with drill progressions and coaching guides, and making it available to programs, trainers, and individual players at every level. The goal is simple. Give every coach and every player access to a tool that makes great shooting form trainable, repeatable, and measurable. The Archer isn't just a product. It's how I coach.